

Revisiting the Slope of the M-dwarf Radius Valley with the 2026 Confirmed-Planet Catalog

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ABSTRACT

The radius valley separating super-Earths from sub-Neptunes provides a sensitive test of planet formation and evolution, but its behavior around M dwarfs remains debated: published slopes ($m = d \log R_p / d \log P$) disagree in sign, and the valley may vanish entirely at the lowest stellar masses. Using the NASA Exoplanet Archive (accessed 2026 July), we construct stellar-mass subsamples of transiting planets and characterize the radius distribution with complementary statistics: a bootstrapped gap-depth measure, Hartigan’s dip test with power simulations, a super-Earth to sub-Neptune count ratio, and two independent slope estimators (a rotating-band density scan and a support-vector-machine boundary fit). We recover the valley in K- and G-type control samples with slopes consistent with published values, validating the pipeline. Around mid-to-late M dwarfs ($M_\star < 0.4 M_\odot$) the radius distribution is unimodal and robust to radius-precision cuts, with super-Earths outnumbering sub-Neptunes. For the combined M-dwarf sample the two estimators return negative slopes ($m = -0.13$ and -0.16) when the fit is confined to the physically plausible valley region, favoring the mass-loss sign, although the uncertainties do not exclude a positive slope at 95% confidence. We further show that an unconstrained gap search instead locks onto the detection-completeness boundary, demonstrating that methodological choices can flip the inferred sign. We suggest this sensitivity contributes to the literature disagreement.

1. INTRODUCTION

The distribution of small-planet radii contains a narrow deficit near $1.5\text{--}2.0 R_\oplus$, the radius valley, that separates rocky super-Earths from gas-enveloped sub-Neptunes (Fulton et al. 2017). Because a planet’s size reflects whether it has retained a hydrogen-helium envelope, the valley is a direct observational probe of planet formation and evolution. Its discovery was enabled by improved stellar radii—first from spectroscopy and subsequently from Gaia parallaxes—which sharpened planetary radii enough to resolve the feature (Fulton & Petigura 2018). The valley is not a fixed radius but shifts with orbital period: around FGK stars it moves to smaller radii at longer period, with a slope $m \equiv d \log R_p / d \log P \approx -0.11$ (Van Eylen et al. 2018).

The sign of this slope discriminates between formation scenarios. Thermally driven atmospheric mass loss—whether photoevaporation by stellar high-energy radiation (Owen & Wu 2017) or core-powered mass loss—predicts a negative slope, because planets closer to the star are stripped to larger core radii. A positive slope would instead point to a valley established at formation in a gas-poor inner disk, without the need for subsequent stripping. Around Sun-like stars the observed negative slope is consistent with the mass-loss picture, although

the negative sign alone does not distinguish photoevaporation from core-powered mass loss.

Whether the same picture holds around M dwarfs, the most common stars in the Galaxy, remains contested. Cloutier & Menou (2020) measured a *positive* slope of $+0.058 \pm 0.022$ for planets around low-mass stars, opposite in sign to the FGK trend, and interpreted it as evidence for an increasing role of gas-poor formation at low stellar mass. In contrast, Van Eylen et al. (2021), using a support-vector-machine boundary fit, found a *negative* slope near -0.11 for M-dwarf planets, consistent with the FGK trend and with mass loss. More recent work suggests the valley may become shallow or disappear entirely toward the lowest stellar masses, leaving the radius distribution effectively unimodal. The disagreement is notable because the studies analyze similar stellar populations yet reach opposite conclusions, raising the possibility that sample selection and methodology, rather than a physical difference, drive the discrepancy.

The number of confirmed transiting planets has grown substantially since these analyses, offering an opportunity to revisit the question with a larger sample. In this work we use the 2026 NASA Exoplanet Archive to characterize the M-dwarf radius distribution and to measure the valley slope with two independent estimators, validating the pipeline against FGK control samples.

Rather than aiming to settle the debate, we ask which sign the current data favor and, equally, how strongly the answer depends on analysis choices. We describe the data in Section 2, the methods in Section 3, our results in Section 4, and their interpretation in Section 5.

2. DATA

We use the NASA Exoplanet Archive Planetary Systems Composite Parameters table (`pscomppars`), which provides one row per planet by combining literature values, accessed 2026 July. We query for transiting planets (`tran_flag` = 1) with non-null planetary radius R_p (`pl_rade`), orbital period P (`pl_orbper`), and host stellar mass $M_\star$ (`st_mass`). Transit-derived radii and periods are the two quantities that define the radius valley: the transit depth sets $R_p/R_\star$, and the interval between transits sets P . Because the radius valley is a narrow feature, the reliability of R_p —which inherits the uncertainty of the adopted stellar radius—is the limiting factor, and we control for it explicitly below.

We restrict the analysis to $R_p < 4 R_\oplus$ and $P < 100$ d. The upper radius limit excludes gas giants, which are unrelated to the super-Earth/sub-Neptune transition, while retaining both flanks of the valley. The period limit excludes the long-period regime where transit detection is highly incomplete. For each planet we compute the fractional radius uncertainty as the mean of the absolute upper and lower errors divided by R_p , and adopt an 8% cut for the fiducial sample, following the precision regime in which the valley was first resolved. We partition the sample into stellar-mass bins approximating spectral type: mid-to-late M ($M_\star < 0.4 M_\odot$), early M (0.4–0.6), K (0.6–0.85), G (0.85–1.1), and F (1.1–1.4 $M_\odot$), plus a combined M-dwarf bin ($M_\star < 0.6 M_\odot$) used for slope measurement, where the individual M subsamples are small. These boundaries are approximate and are tested for sensitivity.

3. METHODS

3.1. Distribution Statistics

To assess whether a subsample is bimodal we use three complementary statistics. First, a gap-depth statistic $D = 1 - \rho_{\text{valley}}/\overline{\rho_{\text{peak}}}$ compares the number density in a valley window to the mean density of the two flanking peak windows, where each density is normalized by the logarithmic width of its window. $D \rightarrow 1$ indicates an empty valley and $D \rightarrow 0$ indicates no depletion. Because D presupposes two populated peaks, we adopt a mass-scaled window centered on the expected valley location $R_{\text{valley}} = 1.86 (M_\star/M_\odot)^{0.18} R_\oplus$ (Petigura et al. 2022) and return D as undefined when either flanking window contains fewer than five planets; this prevents

the statistic from measuring the edge of a unimodal distribution.

Second, we apply Hartigan’s dip test to $\log_{10} R_p$, whose null hypothesis is unimodality. To interpret a non-rejection we estimate the test’s power by generating synthetic bimodal distributions matched to the control-sample size and measuring the fraction in which the dip test rejects unimodality.

Third, as a statistic that assumes no particular modality, we count planets below and above the mass-scaled valley line to form a super-Earth to sub-Neptune ratio. Uncertainties on D and on the ratio are obtained by bootstrap resampling ($N = 2000$) with a fixed random seed.

3.2. Slope Estimators

We model the valley as $R = R_0 (P/10 \text{ d})^m$, a line in the $\log P$ – $\log R$ plane with slope m and reference radius R_0 at $P = 10$ d. We measure m with two estimators of differing principle.

The rotating-band scan searches a grid in (m, R_0) and, for each candidate line, counts planets within a band of half-width $hw = 0.04$ dex and compares this to the count in adjacent reference regions; the line maximizing the deficit relative to its flanks is selected. Grid points whose reference region is sparsely populated are skipped.

The support-vector-machine (SVM) boundary fit labels planets above and below an initial mass-scaled dividing line and fits a linear maximum-margin hyperplane $w_0 \log P + w_1 \log R + b = 0$, from which the slope follows as $m = -w_0/w_1$. Planets are relabeled by the fitted boundary and the fit is iterated to reduce dependence on the initial labels. We use soft-margin strength $C = 1$ and balanced class weights in the fiducial configuration.

For both estimators, confidence intervals are obtained by bootstrap resampling in which period and radius are resampled with the same indices to preserve each planet’s (P, R) pairing. For the M-dwarf sample we additionally restrict to $P < 30$ d to limit contamination from detection incompleteness at long period. We validate both estimators on the G and K samples, where the recovered slopes can be compared to published FGK values.

The hyperparameters above (hw , C , the mass bin edges, and the initial dividing line) are analysis choices rather than measured quantities; we therefore vary each and report the resulting spread in Section 4.5. The valley-location scaling $(1.86 (M_\star/M_\odot)^{0.18})$ is adopted from the literature (Petigura et al. 2022), and the random seed and bootstrap count affect only the stability of the intervals, not the point estimates.

4. RESULTS

4.1. *The Radius Distribution Is Unimodal Around Low-Mass M Dwarfs*

Figure 1 shows the radius distributions of the five stellar-mass subsamples under three radius-precision cuts. The mid-to-late M-dwarf sample ($M_\star < 0.4 M_\odot$) exhibits a single peak near $1.2 R_\oplus$ that persists across all three cuts, from the full sample down to planets with radius uncertainties below 8%. The peak location and unimodal character are therefore not artifacts of measurement precision. In contrast, the G and K subsamples develop an increasingly clear bimodal structure as the precision cut is tightened, with a depletion near $1.8 R_\oplus$ separating the super-Earth and sub-Neptune populations. The early-M subsample ($0.4\text{--}0.6 M_\odot$) is intermediate, showing a partially filled depletion.

4.2. *Failure of One-Dimensional Bimodality Tests*

A fixed-window gap-depth statistic D returns a large positive value for the mid-to-late M sample ($D = 0.41$). Diagnostic counts show that this reflects the sharp upper edge of a unimodal distribution rather than a true valley: the putative upper peak window ($2.2\text{--}2.8 R_\oplus$) contains only 13 planets, so the statistic measures a cliff, not a gap. When the depth statistic is restricted to bins in which both flanking peaks are populated, the mid-to-late M value becomes undefined, consistent with the absence of a valley.

Hartigan’s dip test does not reject unimodality for any subsample, including the G sample in which a valley is visually and statistically evident (dip $p = 0.97$). A power simulation using synthetic bimodal distributions matched to the G sample size recovers the injected bimodality in 93% of trials. The failure to reject unimodality in real data therefore does not arise from insufficient statistical power; rather, the one-dimensional test is degraded because the valley is inclined in the period–radius plane, so that superposing multiple periods fills the depletion. This motivates a direct measurement of the valley slope.

4.3. *The Super-Earth to Sub-Neptune Ratio Increases Toward Lower Stellar Mass*

The ratio of super-Earths to sub-Neptunes, defined relative to a mass-scaled valley location, increases monotonically with decreasing stellar mass, from 0.39 (F) and 0.49 (G) to 0.79 (K) and 1.49 for the mid-to-late M sample (Figure 2). The mid-to-late M value exceeds unity at 1.49 [1.22, 1.83] (68% confidence). Because this statistic makes no assumption about the modality of the distribution, it is not subject to the failure mode of Section 4.2. The trend is consistent with a valley

that fades toward low stellar mass. We caution that these are raw counts without completeness correction; because small planets are harder to detect, the super-Earth counts are underestimated, so the true ratios are likely larger, which strengthens rather than weakens the direction of the trend.

4.4. *Valley Slope: Two Independent Estimators*

We measure the valley slope with two methods of differing principle. The rotating-band scan locates the orientation of the band that is most depleted relative to its flanks; the support-vector-machine (SVM) boundary fit labels planets above and below a mass-scaled dividing line and finds the maximum-margin hyperplane separating them. Applied to the G and K control samples, the band scan returns $m = -0.09$ and -0.07 , respectively. The G-sample value matches the FGK valley slope of $m = -0.09$ reported by Van Eylen et al. (2018), and both control slopes are consistent in sign and magnitude with published FGK measurements near -0.11 , validating the pipeline.

For the combined M-dwarf sample ($M_\star < 0.6 M_\odot$; period restricted to $P < 30$ d to mitigate incompleteness), the band scan returns $m = -0.13$ (95% CI $[-0.23, +0.07]$) and the SVM boundary returns $m = -0.16$ (95% CI $[-0.28, -0.02]$) in its fiducial configuration, with a boundary radius of $1.46 R_\oplus$ consistent with the expected valley location. Both point estimates favor the negative, mass-loss sign, and the fiducial SVM interval excludes both zero and the previously reported positive value ($+0.058$). Our SVM slope is furthermore consistent with the $m \approx -0.11$ obtained by Van Eylen et al. (2021) using the same boundary-fitting technique on an independent M-dwarf sample. Figure 3 shows the two fitted loci overlaid on the M-dwarf sample.

4.5. *Sensitivity to Methodological Choices*

Both estimators show sensitivity to their hyperparameters, which we report rather than suppress. For the band scan, varying the band half-width across 0.03, 0.04, and 0.05 dex yields slopes of $+0.06$, -0.13 , and -0.18 ; the two wider settings agree on a negative slope, while the narrowest is dominated by small-number fluctuations. For the SVM boundary, a scan over the initial dividing line ($\pm 10\%$) and the soft-margin strength C produces nine configurations. Three (all at $C = 0.1$) are rejected by a pre-defined validity criterion ($|m| < 0.5$ and $1.2 < R_0/R_\oplus < 2.2$), because the margin becomes so soft that the boundary crosses the bulk of the distribution at $R_0 \approx 0.3\text{--}0.5 R_\oplus$ rather than tracking the valley. Among the six configurations satisfying the criterion, the slope has a median of -0.08 and ranges over $[-0.23, +0.02]$.

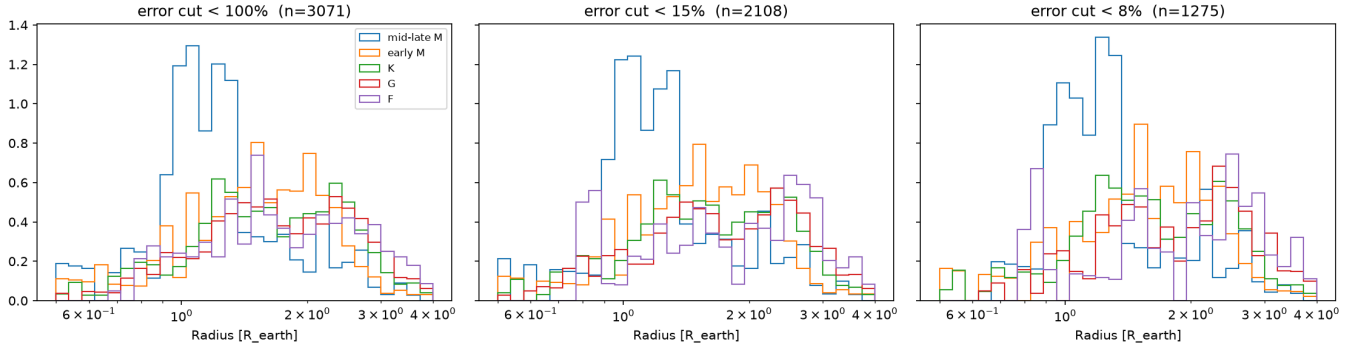


Figure 1. Normalized radius distributions for five stellar-mass subsamples under radius-precision cuts of 100%, 15%, and 8% (left to right). The mid-to-late M-dwarf peak near $1.2 R_{\oplus}$ is stable across cuts, while the G and K depletion near $1.8 R_{\oplus}$ sharpens as precision improves.

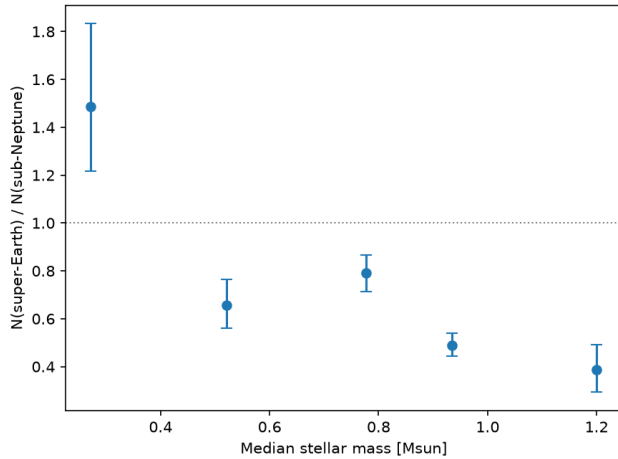


Figure 2. Super-Earth to sub-Neptune count ratio versus median stellar mass for each subsample, with 68% bootstrap intervals. The dotted line marks parity. The ratio rises toward lower stellar mass and exceeds unity for the mid-to-late M sample.

As an extreme illustration of the same effect, an unconstrained gap search (no prior on the slope range) instead converges on a locus of positive slope that traces the lower-right edge of the occupied region—the boundary set by transit detection completeness, where small planets at long periods go undetected—rather than the valley. This confirms that the inferred slope, and even its sign, can depend on how the search is constrained and on the treatment of detection bias.

5. DISCUSSION

5.1. Physical Interpretation

The negative valley slopes recovered for the combined M-dwarf sample by both estimators ($m = -0.13$ and -0.16) are consistent in sign with the value observed around FGK stars and with the prediction of thermally driven atmospheric mass loss, in which planets closer to

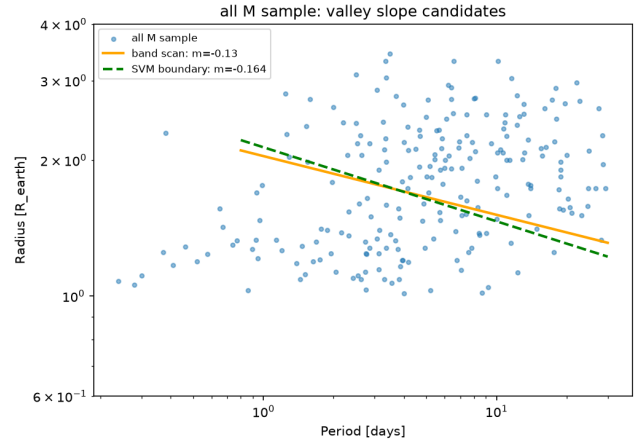


Figure 3. Period–radius distribution of the combined M-dwarf sample with the valley loci from the band scan (solid) and the SVM boundary (dashed). The two independent estimators return consistent negative slopes.

the host star retain their gaseous envelopes less effectively and the valley consequently shifts to larger radii at shorter period. A positive slope, by contrast, would favor a scenario in which the valley is imprinted at formation rather than carved by later mass loss. Our point estimates therefore weigh in favor of a mass-loss origin that operates similarly across stellar mass, rather than a distinct low-mass formation channel. We emphasize, however, that a negative slope alone does not distinguish photoevaporation from core-powered mass loss, since both predict a comparable sign and magnitude; separating these mechanisms requires additional information, such as the dependence on stellar age or insolation, which is beyond the scope of this work.

The unimodal radius distribution of the mid-to-late M sample (Section 4.1) and the super-Earth-dominated count ratio (Section 4.3) are jointly consistent with a valley that becomes shallower toward the lowest stellar masses, to the point where a slope is no longer well de-

finer. This is why we report the mid-to-late M slope as undefined rather than flat: there is no resolved bimodal valley whose inclination could be measured.

5.2. *Why the Literature Disagrees*

Our most robust finding is not a single slope value but the demonstration that the inferred slope, and even its sign, depends on methodological choices. Within a single dataset, varying the band half-width changes the band-scan slope from +0.06 to -0.18 , and an unconstrained gap search converges on the detection-completeness boundary at strongly positive slope rather than on the valley (Section 4.5). Because published analyses differ in sample selection, radius-precision cuts, completeness treatment, and fitting method, we suggest that at least part of the disagreement over the sign of the M-dwarf valley slope reflects this sensitivity rather than a genuine physical difference between samples. In this interpretation, the tension between a reported positive slope and the negative values found by other groups is a symptom of a measurement that is not yet well constrained by the available data.

5.3. *Limitations*

Several limitations bound the strength of our conclusions. First, our counts are not corrected for detection completeness; because small planets are preferentially missed, particularly around larger and more distant stars, raw comparisons across stellar type are qualitative rather than quantitative. We have used the direction of this bias defensively—for example, noting

that completeness correction would only strengthen the super-Earth excess at low stellar mass—but we do not attempt an occurrence-rate calculation, which would require injection-recovery testing beyond the scope of an archival analysis. Second, the combined M-dwarf sample ($n = 225$ after cuts) is small, and the resulting confidence intervals do not exclude a positive slope at 95% confidence in the fiducial band-scan configuration. Third, the F-type results should be treated with caution, since small planets are especially difficult to detect around hot stars and the lower peak of the distribution is likely suppressed by selection effects.

5.4. *Outlook*

The limiting factor is sample size rather than method. Using a pseudo-scaling exercise in which the bootstrap is drawn from an inflated sample of the same distribution, we find that the slope confidence interval narrows substantially as the effective sample grows, while remaining centered on a negative value. We caution that this exercise holds the distribution fixed and therefore underestimates the true uncertainty, since newly discovered planets will populate previously sparse regions of the period–radius plane; it should be read as an indication of the scaling, not a forecast of a specific future interval. A decisive measurement of the M-dwarf valley slope will likely require the larger and more uniform samples of short-period planets expected from ongoing and future transit surveys, together with careful, survey-specific completeness modeling.

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